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An interpretable framework for task response in IELTS writing: integrating thematic analysis, argumentation dimensions, and GPT-4 for educational feedback

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Abstract

The assessment of Task Response (TR) in high-stakes exams like IELTS is plagued by scorer subjectivity, conceptual vagueness, and limited pedagogical utility, which Automated Writing Evaluation (AWE) tools, including “black-box” Generative AI, have not consistently addressed in formative contexts. This challenge is further compounded by variable levels of feedback literacy among learners. To bridge these gaps, this exploratory, proof-of-concept study develops an interpretable, five-dimensional TR framework (Coverage, Opinion, Claim, Grounds, Penalty) through thematic analysis of 28 official IELTS essays (Bands 5.0–7.0). The proposed framework demonstrates promising inter-rater reliability (Cohen’s kappa = 0.827) and strong internal content validity ($S-CVI = 0.92$) within the context of official descriptors. Preliminary results from Kruskal-Wallis ($p < 0.001$) and Spearman correlations ($r > 0.90$ for Claim and Grounds) suggest its potential to discriminate between proficiency levels. By mapping this framework onto GPT-4 via optimized prompts, the study shifts AI’s role from an opaque scorer (Mean Absolute Error, MAE = 0.54; correlation with human scores, $r = 0.39$) toward a structured diagnostic feedback tool (MAE = 0.41; $r = 0.61$). While these results represent a promising proof-of-concept for formative use, the levels of agreement do not yet meet the requirements for high-stakes summative assessment. Given the limited sample size and exploratory design, these findings should be interpreted cautiously as preliminary evidence for framework viability. The study highlights the framework’s potential to enhance pedagogical transparency and support self-regulation through ZPD-aligned interventions. Future research is needed to validate the framework in larger and more diverse samples and to examine its practical impact in classroom settings.

Keywords Automated writing evaluation, GPT-4, IELTS writing, Task response, Thematic analysis

Background

In high-stakes writing assessments such as IELTS, the evaluation of Task Response (TR) remains a persistent and multifaceted challenge, creating a significant gap between testing and teaching. From a cognitive perspective, writing process theory (Flower & Hayes, 1981) conceptualizes writing as a dynamic, recursive problem-solving activity. The quality of a candidate's TR directly reflects higher-order thinking skills, including idea generation, rhetorical decision-making, and metacognitive control. However, official TR descriptors are relatively vague and overlap conceptually with the Coherence and Cohesion (CC) dimension. This vagueness leads to significant examiner subjectivity, unreliable scoring, and ultimately, feedback that fails to guide meaningful student improvement (Shabani & Panahi, 2020; Uysal, 2010).

Compounding this issue from a sociocultural perspective (Vygotsky, 1978) is the critical role of the Zone of Proximal Development (ZPD), where effective learning occurs through targeted scaffolding. Current TR assessments, however, rarely generate the specific, actionable feedback necessary for teachers to provide such support (O'Connell, 2022; Ou et al., 2024). This pedagogical failure can be exacerbated by cross-cultural rhetorical differences. Assessors' expectations of direct, linear argumentation can lead to the misjudgment of effective but culturally distinct writing patterns, highlighting concerns regarding assessment fairness and validity (Dobrić, 2024; Dobrić & Sigott, 2014; O'Grady, 2025).

The rise of Automated Writing Evaluation (AWE) tools promised greater efficiency but has largely failed to address these core issues. Traditional AWE systems focus on surface-level language features (e.g., grammar, vocabulary) while neglecting the higher-order reasoning and argumentation skills central to TR (Karatay & Karatay, 2024; Shermis & Burstein, 2013). Although generative AI (GAI) models like GPT-4 demonstrate superior text comprehension, their "black-box" nature precludes the transparent, interpretable feedback needed to diagnose weaknesses in argumentative logic, thus limiting their pedagogical utility (Yavuz et al., 2024; Zou & Huang, 2024).

Consequently, a fundamental contradiction persists: while TR is a core component of writing proficiency, its assessment lacks a fine-grained, interpretable, and pedagogically applicable framework. This gap has a direct impact on learners. The opaque nature of both human and automated feedback undermines the development of feedback literacy—the capacity of learners to understand, process, and act on feedback to improve their work (Carless & Boud, 2018; Carless & Winstone, 2020). Without a clear framework to make assessment criteria explicit, students struggle to decode evaluative judgments and translate them into actionable steps for improvement.

To address this dual challenge in a structured manner, this exploratory study integrates cognitive and sociocultural theories to construct an interpretable analytical framework as a proof-of-concept. This framework operationalizes TR into five core dimensions (Coverage, Opinion, Claim, Grounds, Penalty). The aim is to explore the potential of redefining TR assessment, moving it towards becoming a more powerful tool for formative feedback and learning development, rather than remaining solely a source of summative judgment. It is important to note that this study is designed as an initial, small-scale investigation. It is important to note that this study is designed as an initial, small-scale investigation. The primary goal is to establish the feasibility and internal

consistency of the framework, not to make broad generalizations about its impact across diverse educational contexts.

Therefore, this study is guided by the following research questions:

RQ1 (Development & Validity): How can a five-dimensional analytical framework for TR be developed and operationalized through thematic analysis, and what is its reliability and content validity?

RQ2 (Discriminatory Power): To what extent does this framework statistically distinguish between IELTS writing bands (5.0–7.0), and what are the key dimensions driving these differences?

RQ3 (AI-Human Alignment): What is the level of agreement between GPT-4 scoring (when guided by this framework) and human expert scoring, and how does prompt optimization based on the framework improve this agreement?

RQ4 (Pedagogical Application): How can this framework be operationalized to support teachers in providing diagnostic, formative feedback in authentic classroom environments?

Literature review

This literature review aims to establish a theoretical and methodological foundation for addressing the longstanding issues in the evaluation of TR in IELTS Writing Task 2. First, it positions academic writing as a cognitive practice situated in social contexts, thereby rendering TR not merely an output but a window into complex literacy activities (2.1). Then, it critically examines the specific assessment challenges arising from this complexity—examiner subjectivity, conceptual vagueness, and cross-cultural biases—and their implications for pedagogical utility (2.2). Subsequently, the review traces how AWE tools have historically sidestepped these higher-order issues in favor of lower-order language features (2.3), with persistent problems in interpretability and equity (2.4), even advanced GAI models struggling to overcome this limitation. Thematic analysis is proposed as a methodological bridge to operationalize TR structures, enabling reliable and transparent evaluation (2.5). The review concludes by integrating these threads into the research gaps that this study aims to address (2.6).

Writing theories: from cognitive processes to Sociocultural practices

Writing assessment has undergone significant paradigm shifts, evolving from behaviorism's focus on surface-level language products to more sophisticated views of writing as both a cognitive process and a sociocultural practice. From the cognitive perspective, writing is conceptualized as a recursive and goal-directed problem-solving activity that engages a range of metacognitive strategies, including planning, translating, and revising (Flower & Hayes, 1981). This model emphasizes the individual writer's mental operations and underscores the importance of strategy use in shaping writing quality.

In contrast, the sociocultural perspective foregrounds the role of social interaction, mediation, and cultural tools in shaping writing development. Rooted in Vygotsky's (1978) theory of learning, this view highlights how progress occurs most effectively within the ZPD, where learners benefit from scaffolded support provided by teachers, peers, or technological tools (Guo et al., 2024; Hyland & Hyland, 2019; Ou et al., 2024). Writing, therefore, is not only an individual cognitive act but also a socially situated practice in which meaning-making is collaboratively negotiated.

Taken together, these perspectives underscore that writing is simultaneously a cognitive problem-solving activity and a socially mediated practice. This integrated theoretical foundation has profound implications for writing assessment: rather than functioning merely as a summative mechanism of judgment, assessment must also serve formative purposes by generating diagnostic, dialogic feedback that guides learner growth and development (Carless & Boud, 2018; Guo et al., 2024; Hattie & Timperley, 2007; Malik & Salman, 2025).

Triple challenges in assessing task response and their pedagogical consequence

The evaluation of TR is fraught with challenges stemming from its inherent complexity, which can be distilled into three interrelated issues:

Examiner subjectivity and low reliability. Judgments of concepts such as “argument sufficiency” are inherently interpretive, leading to significant inter-rater discrepancies (Uysal, 2010; O’Connell, 2022). This subjectivity directly undermines scoring reliability, thereby affecting assessment validity.

Conceptual irrelevance and vagueness. The operational distinction between TR (content relevance and sufficiency) and CC (organizational structure) is often blurred in practice (see IELTS.org). An essay with relevant ideas but poor organization poses a dilemma: Is the flaw in TR or CC? This ambiguity reflects the complex recursive interactions between idea generation and organization in the writing process (Halliday & Hasan, 1989).

Theoretical concerns regarding cross-cultural biases and inequities. Rooted in Kaplan’s (1966) contrastive rhetoric, research consistently shows that rhetorical patterns are culturally shaped. It has been argued that IELTS, in privileging linear, direct argumentation, may potentially penalize writers who favor indirect or circular modes, raising theoretical concerns about the fairness and ecological validity of assessment for global candidate populations (Kobayashi & Rinnert, 2008). While this study acknowledges these concerns as an important theoretical backdrop influencing assessment design, its primary focus is on developing a core analytic framework. The empirical investigation of cross-cultural effects remains beyond the scope of this proof-of-concept study and is identified as a key direction for future research (see Sect. 5.4).

The collective consequence of these challenges is a dual failure: the loss of pedagogical validity and the suppression of feedback literacy. Teachers find it difficult to translate vague band descriptors into actionable feedback that learners can address within their ZPD. More fundamentally, the opaque feedback generated under such conditions fails to develop students’ feedback literacy—the capacity to understand, process, and act on feedback (Carless & Boud, 2018). Without specific, diagnostic guidance, students cannot answer the crucial question “Where to next?” (Hattie & Timperley, 2007), and the formative potential of TR assessment is negated, failing to support learning effectively (Hyland & Hyland, 2019; Panadero & Jonsson, 2013). The framework proposed in this study, therefore, aims to address not only the core assessment challenges of subjectivity and vagueness but also this core pedagogical failure.

The evolution of automated writing evaluation and its persistent limitations

AWE systems were developed to enhance the efficiency and scalability of writing assessment. However, from rule-based systems (e.g., PEG) to statistical models (e.g., e-rater),

and more recently deep learning approaches, their evolution has consistently prioritized the evaluation of lower-level language features (e.g., grammar, vocabulary) over higher-order concerns like argumentation (Shermis & Burstein, 2013; Zou & Huang, 2024).

While these systems can provide reliable scores based on surface features, their feedback is often pedagogically unhelpful. They can identify a text as “weak” but fail to explain where or how to improve—a key flaw known as the “Black-box” problem (Yavuz et al., 2024; Zou & Huang, 2024). As Hattie and Timperley (2007) emphasize, effective feedback must answer three questions: “Where am I going?” (goals), “How am I going?” (progress), and “Where to next?” (actions). Traditional AWE fails on the third, rendering it inadequate for formative pedagogy and limiting its integration into meaningful teaching practices.

Generative artificial intelligence: old problems in new guise

GAI (e.g., GPT-4) represents a potential quantum leap in natural language processing, offering new hope for analyzing complex argumentative structures (Impey et al., 2025). However, it amplifies rather than resolves AWE’s core limitations.

Its sophisticated outputs are generated through inscrutable “Black-box” processes, further exacerbating interpretability issues. Moreover, since GAI models are trained on vast and often culturally biased corpora, they may perpetuate or even amplify existing sociocultural biases, posing serious threats to assessment equity (Cheng & Xu, 2025; O’Grady, 2025). While interactive prompt engineering offers a technical workaround (Su et al., 2023), it does not address the fundamental need for robust operationalization of structures themselves. Without a strong framework to guide it, GAI may produce feedback that is both pedagogically meaningless and inequitable, hindering its responsible application in learner-centered assessment ecosystems.

Thematic analysis as a methodological Bridge

To bridge these gaps, a method is needed that can deconstruct abstract TR structures into analyzable units. Thematic analysis (Braun & Clarke, 2006, 2019), particularly guided by systemic functional linguistics (Halliday & Hasan, 1989), offers a rigorous, systematic approach to identifying and quantifying meaning patterns (or themes) related to argumentation (e.g., claims, reasons, warrants).

This method transcends subjective impressions by converting rhetorical traits into quantifiable data on the frequency, relations, and coherence of thematic elements. While prior studies have employed thematic analysis (Crossley & McNamara, 2010) or computational tools for text analysis (Chen & Cui, 2022; McNamara et al., 2010), their frameworks often lack cultural sensitivity and validation across proficiency levels. Thus, a rigorously developed thematic coding scheme can serve as a crucial “interpretive layer” between human judgment and AI computation, forcing GAI outputs beyond holistic scores toward specific, transparent, and actionable diagnostic profiles.

Synthesizing research gaps and this study’s contributions

This review reveals three interrelated gaps in the existing literature:

Theoretical-applied gap: A disconnect between rich sociocultural-cognitive writing theories and the crude tools available for TR assessment, hindering the application of theoretical insights to practice.

Methodological gap: A paucity of research successfully combining the rigor of qualitative, human-centered methodologies (e.g., thematic analysis) with GAI's computational power to create interpretable and scalable assessment models.

Pedagogical-empirical gap: A severe lack of empirical testing of GAI's capacity to provide equitable and pedagogically useful feedback in TR assessment within authentic cross-cultural educational contexts. The cross-cultural dimension, while theoretically critical, remains particularly underexplored in operational, framework-driven studies.

This study aims to directly address these gaps, with a primary focus on the first two. By constructing a theory-based, fine-grained TR analytical framework and validating it through human-AI collaboration, it will contribute to: (a) theory: refining TR constructs; (b) methods: pioneering a hybrid assessment model; (c) practice: equipping teachers with a diagnostic tool for delivering precise and pedagogically transformative feedback. The framework incorporates a descriptive dimension for cultural rhetorical patterns (see Section "[Theoretical framework and coding manual development](#)") to acknowledge this complexity, laying the groundwork for future research to explicitly examine its moderating effects.

Methodology

Research design

This exploratory study employs a mixed-methods design to develop and validate a fine-grained, interpretable coding framework for TR dimensions in IELTS Writing Task 2. The study analyzes 28 essay samples scoring 5.0–7.0 from the Cambridge IELTS series. The primary objective is to construct a framework theoretically grounded in the concept of the ZPD, providing diagnostic, actionable feedback while ensuring high compatibility with IELTS official standards, clear differentiation across bands, and practical applicability. Given the limited sample size, this study is explicitly positioned as a preliminary proof-of-concept investigation aimed at establishing the framework's validity and reliability, rather than generalizing statistical findings to broader populations. GAI (GPT-4) is used solely as an analytical tool for scoring comparisons under strict human supervision, without model fine-tuning.

Integration of qualitative and quantitative phases

To clarify the mixed-methods logic, the study proceeded through three sequential and integrated phases (see Fig. 1). Phase 1 (Framework Development) employed qualitative thematic analysis of official IELTS descriptors and sample essays to derive an initial coding framework, thereby operationalising abstract Task Response (TR) constructs into measurable analytic dimensions. Phase 2 (Quantitative Analysis) statistically examined this framework's ability to discriminate between proficiency levels, providing preliminary evidence of its discriminatory capacity. Phase 3 (AI–Human Alignment) applied the validated framework to structure AI prompts, evaluating its utility for improving automated formative feedback. In this way, the qualitative phase established the

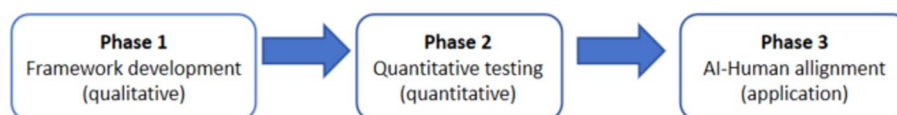


Fig. 1 Sequential mixed-methods design of the study: From framework development to AI application

conceptual foundation, which the subsequent quantitative and application phases tested and operationalised, enabling a progression from theory-building to preliminary validation and practical exploration.

Note: This figure illustrates the study's exploratory sequential mixed-methods design. Phase 1 (Framework development, qualitative) used thematic analysis, iterative coding, and expert validation on IELTS sample essays and official descriptors to derive a five-dimensional Task Response (TR) coding framework (reported with high IRR and content validity indices). Phase 2 (Quantitative testing) applied this framework to $N=28$ curated essays and used non-parametric tests (e.g., Kruskal–Wallis) and correlation analyses to examine discriminatory capacity, yielding preliminary evidence of the framework's ability to distinguish proficiency levels. Phase 3 (AI–Human alignment) operationalised the validated dimensions into structured GPT-4 prompts and compared AI vs. human ratings (reporting MAE and correlation metrics), showing improved alignment after prompt optimisation. This is an exploratory proof-of-concept; interpret results cautiously due to the small, curated sample and the study's developmental scope

Theoretical framework and coding manual development

The coding framework was developed through a rigorous iterative process based on thematic analysis of IELTS official band descriptors, informed by insights from writing process theory (Flower & Hayes, 1981) and sociocultural theory (Vygotsky, 1978). Each category is operationally defined with representative examples to minimize overlap and enhance objectivity.

To acknowledge potential cultural influences on argument presentation, drawing on cross-cultural rhetoric theory (Connor et al., 2008), a “cultural rhetorical patterns” sub-dimension was introduced as a descriptive, non-scoring category. This sub-dimension was designed to quantify organizational strategies employed by writers (e.g., direct/deductive vs. indirect/inductive) and evidence types (e.g., personal anecdotes vs. societal facts) for interpretive context, not to directly influence the TR score or serve as a primary variable for statistical testing in this proof-of-concept study.

Additionally, identified thematic chains (e.g., “advertising → consumer behavior → societal consequences”) serve as references for GPT-4's structured prompt engineering strategies. This design aims to enhance AI's capacity to parse complex arguments by providing explicit, context-based evaluation guidelines. The final five-dimensional coding manual is presented in Appendix A.

Study procedure

The research procedure is divided into three sequential phases:

Phase 1: Framework Development and Validation

An initial coding framework was constructed via thematic analysis. Quality control measures included:

Theoretical consistency: The framework explicitly maps onto cognitive (planning-drafting-revising) and sociocultural (ZPD) models.

Inter-rater reliability (IRR): Two independent coders (PhD candidates in English major with over one year of IELTS teaching experience) coded all texts independently. Cohen's Kappa (via the irr package in R) was calculated across multiple rounds. Following three

rounds of independent coding, calibration, and adjudication, IRR improved from $k = 0.72$ in the first round to $k = 0.83$ in the final independent round.

Content validity: The framework was reviewed and endorsed by three university-level IELTS instructors to ensure relevance and clarity.

Iterative refinement: The coding manual underwent three revisions to enhance clarity and dimensional distinctiveness.

Phase 2: Quantitative Analysis.

Due to the small sample size and non-normal distribution ($n = 28$), non-parametric tests were employed to provide robust preliminary insights without relying on parametric assumptions. Kruskal-Wallis H tests were first used to compare differences in coding dimensions across band scores, followed by post-hoc Mann-Whitney U tests (with Bonferroni correction). Spearman rank correlations and Jonckheere-Terpstra tests assessed monotonic trends between coding scores and proficiency bands. Effect sizes (e.g., eta-squared, ϵ^2) were calculated using the *effectsize* package in R to quantify the magnitude of observed differences. All findings from this phase are interpreted as indicative patterns given the sample constraints.

Phase 3: AI-Human Scoring Consistency.

To evaluate GAI integration, a three-step comparison was conducted:

Baseline AI evaluation: GPT-4 (OpenAI, version September 2024) was used solely under human oversight, without fine-tuning. Initial prompts were straightforward (e.g., “Act as an IELTS examiner and provide a TR score for this essay”). Results were compared to official scores.

Prompt optimization: Initial prompts were refined into structured ones incorporating the coding manual’s dimensions and definitions (e.g., “Using the following TR framework (see Appendix A), analyze this essay. Provide scores for each dimension, an overall TR score, and justifications”). GPT-4 was rerun with optimized prompts.

Consistency verification: Spearman correlations and Cohen’s Kappa values were calculated to measure agreement between baseline/optimized AI scores and human expert scores.

Data collection

Sample selection and rationale

This study utilizes official IELTS sample essays (Cambridge IELTS series 4–18, Bands 5.0–7.0) as the primary data source. The rationale for selecting these samples lies in their unique advantages for TR analysis. Although no isolated TR scores are provided, the accompanying examiner comments explicitly reference TR descriptors to justify overall scores, ensuring transparency and traceability of TR judgments, thereby providing an authoritative benchmark for coding. In contrast, alternative datasets (e.g., learner corpora or classroom essays) often lack examiner validation or dimension-specific feedback.

The focus on the Band 5.0–7.0 range is also pedagogically and empirically motivated. According to the IELTS test taker performance 2022 report (IELTS Partners, 2022, as cited in Koolearn, 2023), the mean writing band score of test-takers worldwide typically falls within this interval, making it the most representative proficiency band for large candidate populations. Targeting this range thus allows the study to capture the majority

of performance variance relevant to TR coding, while avoiding the ceiling and floor effects that may arise from very high or very low proficiency samples.

By relying on these examiner-verified samples, the coding process in this study aligns systematically with TR criteria, minimizing subjectivity and enhancing construct validity. Notably, for half-band scores (e.g., 5.5), the study adopts the holistic band score for TR dimensions, as IELTS official descriptors provide descriptions only for whole bands. Essay distribution is detailed in Table 1.

Acknowledging Sample Size Limitations: The sample size ($N=28$) is appropriate for an exploratory, proof-of-concept study aimed at framework development and initial validation, as recommended for thematic analysis (Braun & Clarke, 2006). However, it limits the generalizability of statistical findings. A post-hoc power analysis using G*Power indicated that a sample size of approximately 30 would be required to detect a large effect ($f=0.4$) with 80% power at $\alpha=0.05$, suggesting our sample is at the lower bound for robust parametric analysis. This justifies our use of non-parametric tests and positions the study as a foundational step for future large-scale validation.

Data processing workflow

A standardized data processing workflow was implemented to ensure reproducibility and rigor:

- **Data Entry and Anonymization:** Essays were transcribed verbatim into plain text files, removing any identifying information.
- **Qualitative Coding:** The coding was conducted in NVivo 20. Two independent coders, following the finalized codebook (Appendix A), performed double-blind coding on the entire dataset. A three-level node structure mirrored the main dimensions and sub-dimensions.
- **Reliability Calibration:** The coders underwent three rounds of independent coding and calibration meetings to discuss discrepancies and refine the codebook. Inter-rater reliability (IRR) was calculated after each round until a consensus was achieved.
- **Statistical Analysis:** All quantitative analyses were performed in R (v4.4.3). The tidyverse package (v1.1.3) was used for data wrangling, ggplot2 (v3.3.4) for visualization, irr (v0.83.1) for IRR analysis, and coin (v1.3.2) for non-parametric hypothesis testing.

Research tools and technical standards

Software configuration

Qualitative analysis (NVivo 20): Maintained a hierarchical coding structure. Matrix coding queries were used to ensure consistency across the project.

Table 1 Sample distribution across IELTS bands

Band	Representative Examples	Number
5	C7-T4, C11-T2, C12-T2, C12-T4	4
5.5	C8-T2, C10-T4, C11-T4, C14-T3	4
6	C5-T3, C12-T1, C13-T3, C13-T4, C15-T2, C16-T1, C17-T4	7
6.5	C8-T4, C13-T1, C15-T4, C17-T1, C17-T2, C17-T3	6
7	C4-T3, C11-T3, C13-T2, C14-T1, C15-T1, C15-T3, C16-T3	7

Cambridge (C), Text (T); e.g., C7-T4=IELTS Cambridge 7, Test 4;

Quantitative analysis (R 4.4.3): Employed the tidyverse ecosystem for data cleaning and transformation. Specific packages for statistical tests are as noted above.

AI-Assisted evaluation system

The GPT-4 model (OpenAI, September 2024 version) was accessed via the official API. All prompts were designed to be structured and reproducible; Appendix C provides examples.

Quality control and ethics

Data security: All data processing was conducted on secure local computing infrastructure.

Copyright compliance: Use of Cambridge IELTS materials adheres to fair use principles for academic research and critical analysis.

AI ethics and disclosure: GAI was employed solely as a research tool for scoring under direct human supervision. It did not contribute to study conceptualization, analysis, or manuscript writing but enhanced text readability.

Funding and conflicts of interest: This study received no specific funding from any agency. The authors declare no conflicts of interest.

Results

RQ1: What are the statistical differences in the five coding dimensions across various band scores (5–7) in exemplary essays?

Addressing this question involved two key steps: clarifying the construction and reliability of the thematic analysis coding framework, and examining key characteristic differences among essays scoring bands 5.0 through 7.0 via statistical analysis.

Construction of the coding framework and its validity and reliability analysis

First, based on Braun and Clarke (2006) thematic analysis method, a systematic coding of discourse features in IELTS Writing Task 2 essays scoring 5.0–7.0 was conducted. The coding process was grounded in IELTS official band descriptors, aiming to capture subtle differences across band scores.

Following iterative development, five core dimensions were finalized: Coverage, Opinion, Claim, Grounds, Penalty Dimension (for capturing negative factors such as insufficient word count, format errors, and excessive repetition). These five dimensions constitute a complete TR evaluation process: content completeness → opinion quality → argumentative logic (Claim & Grounds) → formal accuracy. Each main dimension is further subdivided into 1–3 sub-dimensions (e.g., Opinion includes consistency and clarity), scored on a three-level scale (0–2) corresponding to “insufficient → partially met → fully met”.

Consistent with the theoretical considerations outlined in Section “[Triple challenges in assessing task response and their pedagogical consequence](#)”, cross-cultural rhetorical differences were treated as a moderating variable for descriptive purposes. They are theorized to primarily influence the presentation of Opinion and the organization of Grounds, and were annotated separately rather than being directly incorporated into the TR scores (see Fig. 2). For specific interpretations, refer to Appendix A.

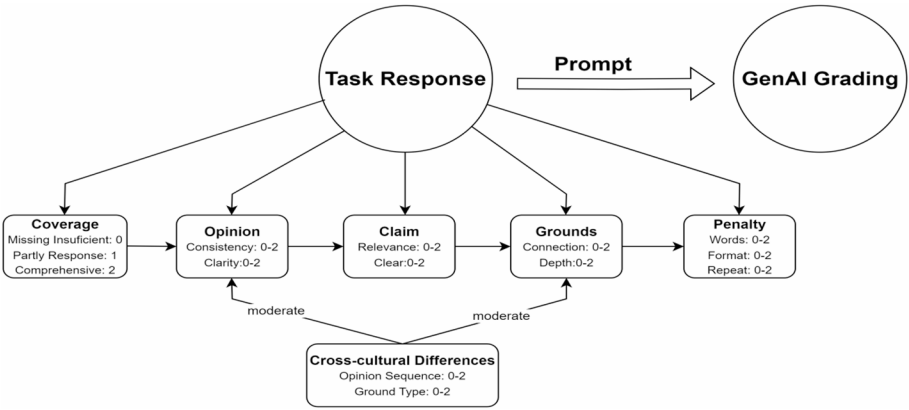


Fig. 2 An operational assessment framework for the dimension of TR

Table 2 Iterative refinement of the coding framework

Round	Average Cohen's Kappa	Key Change	Impact on Discriminatory Power (Kruskal-Wallis <i>p</i> -value)
1	0.72	Initial codebook application	$P = 0.01$
2	0.78	Clarification of Grounds-Depth criteria	$P < 0.005$
3	0.827	Operationalization of "Opinion-Consistency"	$P < 0.001$

Second, two independent coders applied this framework to 28 IELTS sample essays. Inter-rater reliability analysis showed an average Cohen’s Kappa coefficient of 0.827, indicating high consistency within this exploratory study. This high consistency was achieved after three rounds of iterative refinement (see Table 2). Subtle differences in interpreting implicit logic were resolved through expert adjudication, further enhancing coding reliability.

Third, three professional experts independently reviewed the coding framework. The item-level content validity index (I-CVI) was 0.94, and the scale-level content validity index (S-CVI) was 0.92, both far exceeding the standard threshold of 0.78, indicating that the framework aligns highly with IELTS band descriptors in the judgement of these experts.

Finally, after three rounds of iterative optimization (summarized in Table 2), the average score for the “Grounds depth” sub-dimension showed a significant increase in difference between Band 5 ($M = 0.8$) and Band 7 ($M = 2.4$), $p < 0.001$. S-CVI improved from 0.88 to 0.92. The *p*-value from Kruskal-Wallis tests, which provides preliminary support for the discriminatory power of the framework, decreased significantly from 0.01 to < 0.001 (as shown in Table 2), confirming the effectiveness of the optimization process within this dataset.

As shown in Fig. 2, the operationalized TR framework decomposes abstract band descriptors into five measurable indicators plus one moderating variable, establishing a theoretical basis for designing GAI-based scoring prompts.

Performance differences across band scores on TR dimensions

Figure 3 displays the mean change trends (upper panel) and standard deviation distributions (lower panel) of the five core dimensions (Coverage, Opinion, Claim, Grounds, Penalty) across different IELTS Writing band scores (5.0–7.0). Overall, within this ample, the means of each dimension significantly increase with rising band scores, providing initial evidence that the proposed framework can effectively capture fine-grained differences between essays at different levels.

The results suggest that Coverage gradually approaches completeness as band scores increase, with Band 7 essays achieving nearly full coverage. Opinion clarity appears to improve notably beyond Band 6, stabilizing at a high level in Band 7. Claim and Grounds show the most pronounced improvements, emerging as potential key differentiators for mid-to-high scoring essays. Penalty factors decrease significantly with rising band scores, with Band 7 essays largely avoiding issues like word count deficiency and formatting errors. Cross-cultural differences did not directly affect TR scores but showed a potential moderating effect on claim and ground organization around Band 6.5 in this limited sample.

In summary, the proposed TR framework shows promise in distinguish essay levels across Bands 5.0–7.0 within the context of this study, particularly in capturing argumentative depth and logical development, suggesting diagnostic value.

RQ2: How does this coding framework perform in distinguishing different band scores and in terms of internal consistency?

Difference analysis

Due to the limited sample size ($n = 28$), non-parametric statistical methods were adopted to ensure robustness. Kruskal–Wallis H test results indicated significant differences across all five coding dimensions between band scores ($H(4) > 18.0$, $p < 0.001$). It should be noted that while statistically significant, these results are based on a small, curated sample. Post-hoc Mann–Whitney U tests with Bonferroni correction revealed that while differences between adjacent half-bands (e.g., 5 vs. 5.5) were not always significant, the

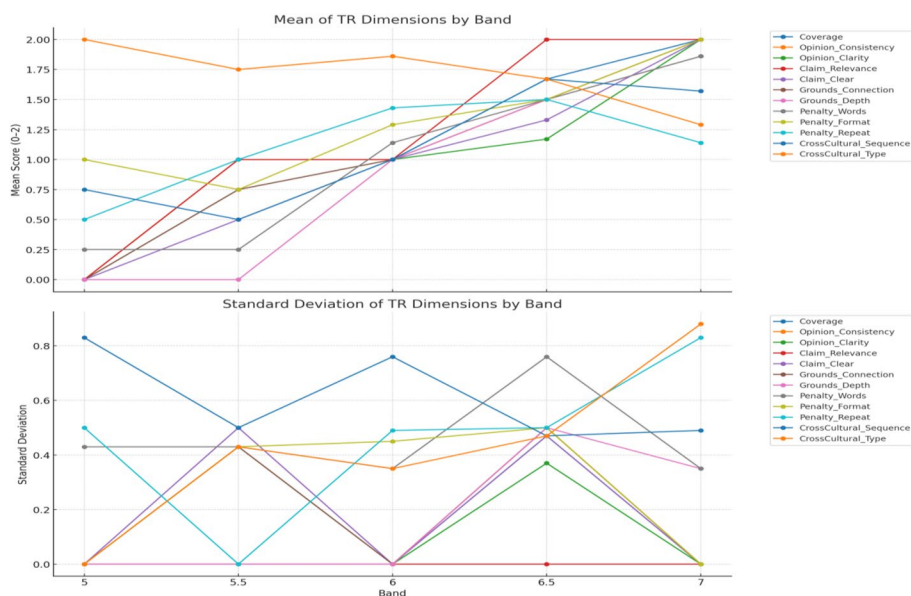


Fig. 3 Differences in TR dimensions across bands

distinctions between full bands, particularly between Band 6 and Band 7, were stark ($p < 0.01$ for Claim and Grounds).

Correlation analysis

As visualized in Fig. 4, “Claim Relevance” ($r = 0.94$), “Coverage” ($r = 0.91$), and “Evidence Depth” ($r = 0.90$) exhibited the strongest correlations with the overall band score within this dataset. The Jonckheere-Terpstra test confirmed a significant monotonic increasing trend for these dimensions across the proficiency bands (Relevance: $JT = 274$, $p < 0.001$; Coverage: $JT = 270$, $p < 0.001$; Depth: $JT = 267$, $p < 0.001$), suggesting an association higher band scores are associated with systematically higher scores on these core argumentation dimensions in the analyzed sample.

The inter-correlation matrix (Fig. 5) further provided support for the internal structure of the framework, showing strong relationships between conceptually linked dimensions (e.g., Opinion Clarity and Evidence Coherence, $r = 0.90$). In contrast, the “Format” dimension had the lowest correlations ($r = 0.49$ – 0.65), implying limited predictive validity in distinguishing band scores in this preliminary analysis.

Overall, the thematic analysis coding framework demonstrated an ability to stably distinguishes essays across band scores in this exploratory study and revealed close relationships between core dimensions and band scores, providing a solid foundation for reliability and consistency testing.

RQ3: What is the consistency between GPT-4 scoring based on this framework and human scoring, and in which dimensions are there significant deviations?

To evaluate GAI logical argumentation optimization, we compared: (1) IELTS examiner vs. baseline GPT scoring, and (2) baseline GPT vs. optimized prompt GPT scoring. The baseline prompt drew on prompt engineering suggestions from Giray (2023), using only official IELTS TR descriptors under a “black-box” approach. The optimized prompt incorporated the five-dimensional framework and 10 detailed sub-dimensions, requiring explicit scoring, textual evidence, and pedagogically relevant feedback. This side-by-side design allowed direct comparison of GAI performance under implicit versus operationalized guidelines. Prompt design differences are summarized in Table 3; full specifications are in Appendix B.

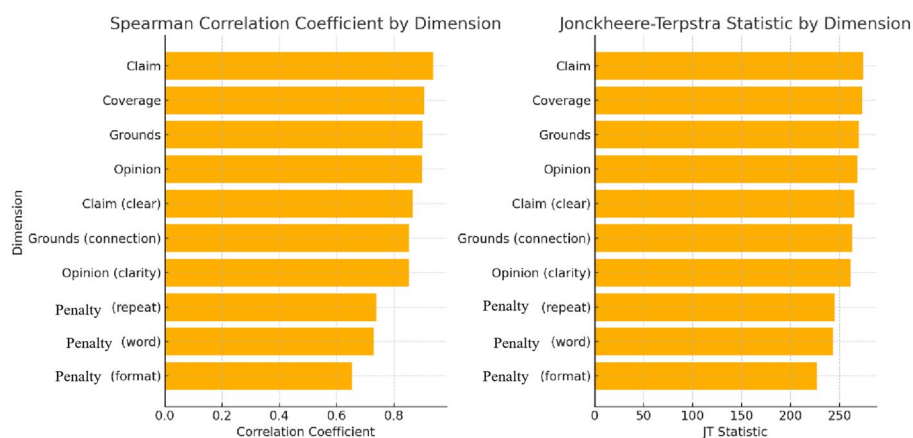


Fig. 4 Spearman correlation coefficient ranking of each dimension and Band

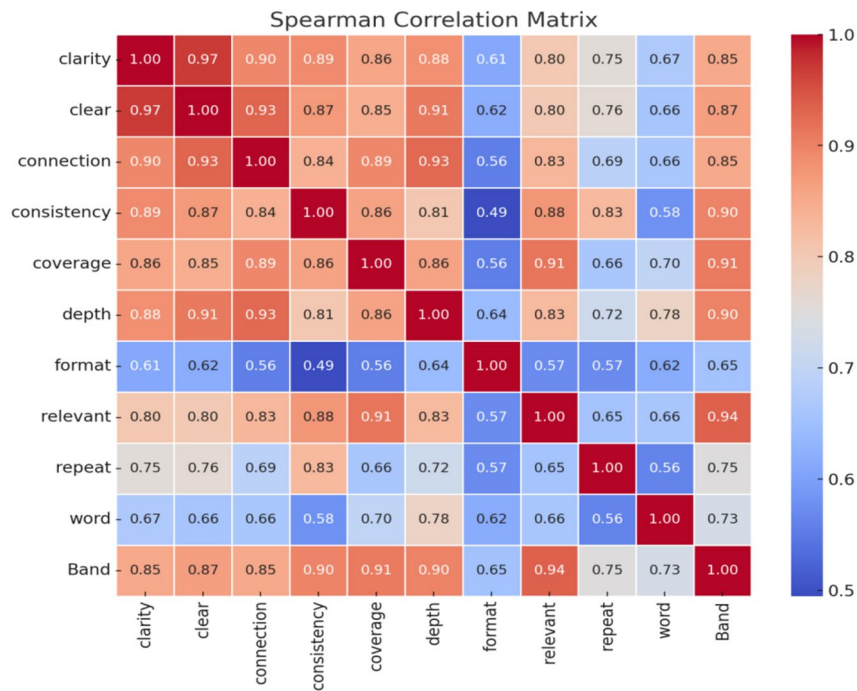


Fig. 5 Spearman correlation matrix (heatmap visualization)

Table 3 Comparison between baseline and optimized prompts for GAI evaluation

Category	Baseline Prompt (Black-box)	Optimized Prompt (Framework-based)
Examiner role	Please act as an official IELTS examiner	Same as baseline, to ensure comparability.
Scoring Criteria	IELTS TR official descriptors	TR framework with 5 dimensions & 10 sub-dimensions, explicitly operationalized
Scale	0–9 IELTS band score	0–2 per sub-dimension, aggregated into IELTS 0–9 band (±0.5 margin)
Evaluation Process	Overall judgment of TR, with general comments	Stepwise scoring for each sub-dimension (Coverage, Opinion, Claim, Grounds, Penalty), with textual evidence
Output Format	Narrative feedback, free text	Structured table: Sub-dimension
Cultural Factor	Not considered	Annotated separately (e.g., opinion sequence, ground type), excluded from TR score
Pedagogical Affordance	Limited diagnostic value; holistic score and overwhelming feedback	Fine-grained diagnostic feedback, aligned with teaching purposes (actionable suggestions for learners)

As shown in Fig. 6, comparisons were made among baseline prompt GPT, optimized prompt GPT, and human examiner scores. During data processing, GPT score ranges (e.g., 4.5–5) were converted to midpoints (e.g., 4.75) to ensure comparability.

Figure 6 displays the comparative analysis of baseline and optimized prompts against IELTS official scores. The baseline prompt had an average absolute deviation of 0.54 from official scores, with relatively weak correlation ($r=0.39$). In contrast, the optimized prompt reduced the average error to 0.41 and significantly improved correlation with official scores ($r=0.61$). These results indicate that the optimized prompt not only aligns more closely with human examiner scoring in this limited comparison but also exhibits higher stability across different band scores. This improvement is particularly evident in mid-level scoring (Bands 6–6.5.5), where the baseline prompt often overestimates

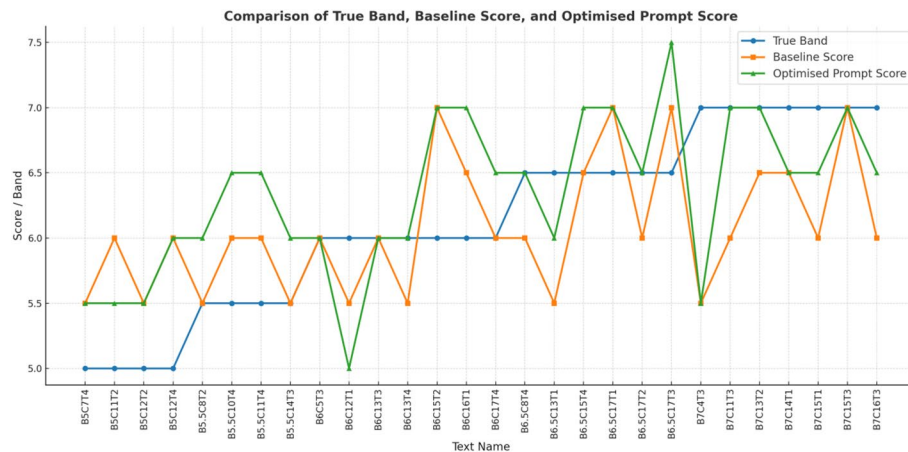


Fig. 6 Score comparison between GPT (baseline & optimized prompt) and IELTS examiners

student performance, while the optimized prompt generates more conservative and accurate evaluations within the constraints of this study.

RQ4: how does this framework support teachers in providing diagnostic feedback in classroom environments?

This section outlines a proposed three-step process for applying the TR framework to generate diagnostic feedback, illustrated with a single case:

Step 1: Dimension-level scoring. Essays are coded across the five dimensions (Coverage, Opinion, Claim, Grounds, Penalty), highlighting strengths and weaknesses per sub-dimension.

Step 2: Identification of gaps. Low scores on specific dimensions pinpoint weaknesses—for example, a Coverage score of 0 indicates unaddressed task components, while a low Grounds score reflects shallow reasoning.

Step 3: Feedback generation. Once weaknesses are located, the coding scheme can be operationalized into structured prompts for feedback. Optimized GPT-4 prompts (Appendix B) require outputs in a table format that links each sub-dimension with (a) the assigned score, (b) textual evidence from the essay, (c) justification for the score, and (d) targeted suggestions for improvement.

To illustrate this process, Appendix C provides a worked example based on a “Band 6” essay from Cambridge IELTS 17 (Test 4, pp. 138). The task prompt contained four elements: (a) people with health problems, (b) trying alternative medicines and treatments, (c) instead of visiting their usual doctor, and (d) evaluation as a positive or negative development.

In this case, the candidate addressed medicines and treatments but neglected the reference to visiting a usual doctor. Accordingly, Coverage was coded as 1. Notably, the official IELTS examiner’s comment stated that the response “addresses both sides of the question”, indicating a divergence between holistic examiner judgment and the fine-grained scoring of the framework. A complete coding of all five dimensions for this sample is provided in Appendix C.

These quantitative improvements in AI alignment provide a basis for exploring the framework’s qualitative pedagogical applications, as discussed in Section “[Discussion](#)”.

Discussion

This study, by developing and validating a five-dimensional TR scoring framework, provides preliminary evidence of its effectiveness in distinguishing IELTS writing levels and enhancing GAI scoring consistency. However, how do these quantitative results translate into educational implications? To deepen the interpretation of the findings, this study draws on Panadero and Jonsson's (2013) framework on the formative use of scoring rubrics, discussing from four dimensions: mechanisms, moderating factors, AI role positioning, and teacher adaptability. The following discussion interprets the findings within the acknowledged boundaries of this exploratory, proof-of-concept study.

Formative role of scoring rubrics: interpreting key findings

The potential effectiveness of the TR analytic scoring framework proposed in this study is suggested by the quantitative results of distinguishing writing levels (RQ1) and improving AI scoring consistency (RQ3). More importantly, it offers a potential pathway for achieving formative assessment. Panadero and Jonsson's (2013) mechanisms offer a powerful theoretical lens for interpreting these preliminary findings.

Enhancing assessment transparency and feedback quality: Breaking down the vague TR construct into five core dimensions (e.g., claim clarity, grounds depth) essentially makes assessment criteria explicit. This could transform feedback from generic judgments like “insufficient task response” into specific diagnoses such as “clear claims but failing to provide in-depth evidential support for core claims”. When injecting this framework into GAI via prompt engineering, we were able to convert a “Black-box” scoring tool into a diagnostic assistant capable of providing interpretable, actionable feedback within the constraints of this experiment, directly addressing longstanding deficiencies in AWE systems.

Reducing anxiety through predictability and supporting self-regulation: The framework demonstrated high reliability and score differentiation in this controlled analysis. More importantly, it could provide learners with cognitive tools for self-regulated learning. Students might plan based on dimensions before writing and self-assess accordingly afterward, a process that could potentially enhance their metacognitive abilities and self-efficacy, promoting a shift from passive test-takers to active writers. However, these proposed benefits remain to be empirically tested in actual learning environments.

Understanding the validity boundaries of the framework: perspective on moderating factors

The full realization of the framework's utility is moderated by a series of factors, and recognizing these boundary conditions is crucial for assessing its claimed scope of validity rather than assuming unrestricted generalisability. Importantly, the validity evidence presented in this study should be understood as interpretive and content-aligned, situated within an exploratory, proof-of-concept design, rather than as predictive or criterion-related validity in a psychometric sense.

Integration with metacognitive activities is key. The framework itself is static; its efficacy would likely need to be activated through dynamic metacognitive practices such as rubric-based self-explanation, peer review, and reflective journaling. Prior research shows that such activities can foster students' self-regulation and internalization of assessment criteria (Panadero & Jonsson, 2013; Ou et al., 2024). Teachers play a vital role

as designers of learning activities and interpreters of feedback, helping students use the framework to diagnose their own issues rather than passively accepting scores.

Attention to washback effects and AI-related impacts is also essential. A clear framework can potentially positively guide teaching and learning, but if misapplied, it may narrow students' writing styles, encouraging rubric-focused compliance at the expense of creativity and individual voice (Zou & Huang, 2024). Similar concerns have been noted in English for Academic Purpose research on assessment washback (O'Connell, 2022) and more recently in studies of AI-mediated feedback in writing classrooms (Malik & Kalim, 2025; Ou et al., 2024; Guo et al., 2024). Accordingly, GAI-generated feedback informed by the framework should not be treated as definitive or authoritative scoring, but as provisional diagnostic input that requires pedagogical interpretation and contextualization by teachers, in order to mitigate unintended negative impacts of technology on learning (Guo et al., 2024; Malik & Salman, 2025). A further consideration concerns the divergence between analytic and holistic assessment. As illustrated in Section "RQ4: how does this framework support teachers in providing diagnostic feedback in classroom environments?" and Appendix C, the framework coded Coverage as 1 because the candidate neglected reference to "visiting their usual doctor", whereas the IELTS examiner's holistic comment described the essay as addressing both sides of the question. This discrepancy underscores the framework's stricter, more fine-grained analytic orientation compared with examiners' broader interpretations. Such divergence does not necessarily indicate instrument error; rather, it highlights a critical validity boundary. While analytic precision can surface subtle task omissions that are pedagogically valuable for formative feedback, holistic scoring tends to tolerate these omissions in favor of overall communicative effectiveness. However, an overreliance on analytic dimensions risks reifying minor omissions as major deficiencies if feedback is not contextualized within broader task expectations. This is particularly salient for learners from non-linear rhetorical traditions (e.g., East Asian writing conventions), theorizing that analytic scoring may further amplify perceived gaps in coherence or sequencing, necessitating culturally sensitive adaptations in feedback practices (Cheng & Xu, 2025; Connor et al., 2008). This study's inclusion of a descriptive cross-cultural dimension is a first step in acknowledging this complexity, but its practical implications for fairness require dedicated future research.

Contextual adaptability further moderates practical outcomes. The framework's effectiveness depends on factors such as educational setting, language proficiency, and cultural rhetorical conventions (as captured in the "cross-cultural differences" dimension). For example, this study's correlation analysis (Fig. 5) showed strong interdependencies between Claim and Grounds ($r=0.93$), suggesting that unaddressed metacognitive gaps could exacerbate washback effects. Teachers are thus encouraged to adapt the framework flexibly to learner profiles and instructional purposes, ensuring that its use supports educational equity and writing development rather than constraining them. These contextual factors intersect with broader moderators discussed in Section "Limitations and future directions", such as socioeconomic and educational backgrounds.

Implications: toward a Human-AI collaborative formative assessment paradigm

The results of this study have potential implications for teaching and AWE development, contingent on further validation.

Potential implications for teachers: Teachers could potentially utilize such frameworks to provide scaffolded feedback within the ZPD and design activities to promote student internalization of assessment criteria. Teachers' professional judgment plays an irreplaceable core role in interpreting AI feedback, designing interventions, and mitigating possible negative washback effects (Guo et al., 2024; Malik & Salman, 2025).

Potential implications for AWE development: GAI's ultimate role in education should not be as an "examiner" but as a "tutor". AWE system design goals should shift from pursuing high consistency with human scoring to generating multi-dimensional, diagnostic formative feedback to support personalized learning. This will transform technology from a judgmental endpoint to a developmental starting point.

Limitations and future directions

This study has several limitations that also chart a course for future research. First, the sample size ($N=28$) and restriction to Bands 5.0–7.0, while justified for this exploratory phase, necessitate caution in generalizing the findings. Given the relatively small dataset and the number of analytical comparisons conducted, the results should be interpreted as indicative patterns rather than as statistically robust or generalizable evidence. Future work must include larger, more diverse corpora encompassing the full score range (Bands 4.0–9.0) and essays from test-takers of diverse linguistic backgrounds to allow for more stable statistical inferences and stronger validity claims.

Second, the study focused exclusively on TR. Writing proficiency is multifaceted, and isolating TR limits the extent to which conclusions can be drawn about overall writing quality or score performance. Future research should therefore investigate the interplay between improvements in TR and other scoring criteria like Coherence and Cohesion, Lexical Resource, and Grammatical Range and Accuracy. Such work could clarify whether gains in argumentative structure are independent of, or interact with, linguistic complexity and discourse-level control, thereby refining the pedagogical interpretation of analytic feedback.

Third, the effects of rubric-based formative assessment are not automatic but are moderated by a host of factors (Panadero & Jonsson, 2013). The present study did not directly measure learners' engagement with feedback, changes in metacognitive awareness, or instructional practices accompanying framework use, which constrains conclusions about learning processes and outcomes. Future studies should employ design-based or classroom-based research methodologies to examine how factors such as learners' metacognitive skills, educational background, cultural rhetoric, and teachers' scaffolding practices shape the framework's impact in authentic instructional settings. In particular, the role of cross-cultural rhetorical patterns, which this study incorporated as a descriptive variable, should be explicitly investigated as a moderating factor in both human and AI-assisted assessment. Such investigation is essential for distinguishing between the framework's analytic potential and its realized pedagogical effects, as well as for developing culturally sensitive adaptations that promote educational equity.

Conclusion

This study builds upon a proof-of-concept exploration to propose a cautiously framed conceptual shift in how we approach writing assessment. By integrating a theory-informed, granular TR framework with the analytical power of GAI, we offer preliminary

evidence for the viability of bridging aspects of standardized evaluation and individualized writing development. The preliminary findings suggest that such a framework has the potential to make assessment more transparent, diagnostic, and ultimately, more pedagogical, pending further validation in larger-scale studies.

However, technological tools are not universal remedies. Their successful implementation would hinge on a collaborative human–AI paradigm, the contours of which are sketched but not finalized by this research. In this proposed vision, AI could handle data-intensive processing and generates provisional diagnostic insights, while teachers provide the essential educational meaning, emotional support, and professional judgment to interpret and act on these insights. From a feedback literacy perspective, this collaborative model foregrounds learners’ ability to interpret and make sense of scoring criteria, thereby aligning with the “appreciating feedback” dimension rather than merely emphasizing passive score reception(Carless & Boud, 2018). By leveraging the complementary strengths of both, writing instruction could move toward practices that are more transparent and pedagogically supportive, fostering learner development without displacing human judgement or high-stakes assessment standards.

In sum, this exploratory study develops and provides initial support for an interpretable TR framework, demonstrating its potential to enhance both the consistency of AI-generated feedback and the clarity of diagnostic assesment. The path from this promising proof-of-concept to a validated pedagogical tool requires further empirical work, particularly to test its effectiveness across diverse cultural and educational contexts and to define the optimal roles for teachers and AI within the formative assessment process.

Appendix A

Table 4 The task response codebook for IELTS

The Task Response Codebook for IELTS		
Dimension	Sub-dimension	Operational Definition
Coverage	Coverage	0=The main parts of the prompt are incompletely addressed. Key requirements may be neglected or misunderstood.
		1=The main parts of the prompt are addressed, but unevenly. Some aspects may be more fully covered than others, or parts of the response may be tangential.
Opinion	Consistency	2=All parts of the prompt are appropriately and sufficiently addressed, with balanced attention to each task requirement.
		0=A position is expressed but is ambiguous or fluctuates. The development of the stance is unclear.
	Clarity	1=A relevant position is presented but may lack full consistency. Conclusions may be unclear, unjustified, or repetitive, weakening the overall stance.
		2=A clear and consistent position is maintained throughout the response, from introduction to conclusion.
		0=The viewpoint is vague or implicit, making it difficult to identify the writer’s definitive stance.
		1=A relevant viewpoint is presented but with limited elaboration or precision. Its position may not be forcefully stated.
		2=The viewpoint is clearly and explicitly stated, and is effectively elaborated upon in the response.

Table 4 The task response codebook for IELTS

The Task Response Codebook for IELTS

Claim	Relevance	0=Main ideas are limited in relevance. Some may be off-topic or insufficiently linked to the prompt's core question. 1=Main ideas are generally relevant to the prompt, though some may lack full development or precise focus. 2=Main ideas are clearly relevant and directly address the prompt. They form a strong foundation for the argument.
	Clear	0=Claims are vague, implicit, or difficult to distinguish from supporting details. 1=Claims are identifiable but may lack sharp focus or precision. The line of reasoning is apparent but could be clearer. 2=Claims are clearly and explicitly stated, establishing a generally coherent and logical line of reasoning.
Grounds	Connection	0=Support is underdeveloped, irrelevant, or poorly connected to the main claims. May include unnecessary details. 1=Some support is relevant, but the connection to claims is not fully developed or is only loosely signaled. Support may feel inadequate. 2=Supporting ideas are clearly connected to the main claims. The relationship between ideas is logical, though links may occasionally be implicit.
	Depth	0=Lacks analysis. Presents statements or facts without explanation or logical development. 1=Provides partial or surface-level analysis. Explains the "what" but limited on the "why" or "how," without extending ideas significantly. 2=Develops ideas with thorough analysis, showing clear logical progression. May occasionally exhibit over-generalisation or a lack of precise focus, which prevents a higher band score.
Penalty	Words	0=Under 250 words, or severe imbalance makes parts of the argument undeveloped. 1=Meets the word count but with uneven development (e.g., one paragraph overly long, another too brief). 2=Meets the word count with balanced development across sections and no major redundancy.
	Format	0=Paragraphing is disorganised; may be missing a clear introduction, body, or conclusion. 1=A basic essay structure is present but may have inconsistencies or awkward breaks in coherence. 2=Clear and logical paragraphing is used throughout, enhancing the coherence of the response.
	Repetition	0=Noticeable repetition of the same ideas. 1=Some repetition of ideas or phrasing occurs. 2=Little to no repetition; ideas progress without redundant statements.
Cross-cultural Difference	Opinion Sequence	0=Indirect: Background or others' views are presented first; the writer's main stance appears late. 1=Mixed: Some indication of stance in the introduction, but it is strengthened or fully clarified later. 2=Direct: The main stance/argument is clearly presented in the introduction (e.g., first or second sentence).
	Grounds Type	0=Primarily personal examples (e.g., "In my experience...", "My friend said..."). 1=Mixed use of personal anecdotes and broader societal evidence or general observations. 2=Primarily societal evidence (e.g., trends, statistics, studies, policies) with minimal or no personal anecdotes.

Cross-cultural difference is included for descriptive and interpretive purposes only. It is annotated separately and does not contribute to the calculation of the overall TR score in this study

Appendix B. Prompts (baseline and optimized prompt)

Baseline prompt: Please act as an official IELTS examiner and assess the following IELTS essays on the Task Response (TR) dimension in strict accordance with the IELTS Writing Task Response scoring criteria.

The goal is to produce a score that is similar to the actual score given by IELTS examiners (± 0.5 margin of error).

Please provide: 1. A single TR band score (0–9). 2. A brief justification for the score, using clear and objective language. 3. General suggestions for improvement.

Optimized prompt: Please act as an official IELTS examiner and assess the following IELTS essays on the Task Response (TR) dimension in strict accordance with the IELTS Writing Task Response scoring criteria.

The goal is to produce a score that is similar to the actual score given by IELTS examiners (± 0.5 margin of error).

In the evaluation process, use the following operationalized Task Response framework: Main Dimensions (5) and Sub-dimensions (10):

1. Coverage – task completion
2. Opinion – consistency, clarity
3. Grounds – connection, depth
4. Penalty – word count, format, repetition

Scoring scheme:

- Each sub-dimension is rated on a 0–2 scale (0 = insufficient, 1 = partially meets, 2 = fully meets).
- Provide “textual evidence” from the essay for each sub-dimension.
- Give a brief justification of why the score was assigned.
- Offer “targeted suggestions” for improvement at the sub-dimension level.
- At the end, provide an “overall TR band score (0–9)” consistent with IELTS, within ± 0.5 margin of error.

Important:

- Do not skip any sub-dimension.
- Use structured, explicit language.
- Present the results in a table format: Sub-dimension | Score | Evidence | Justification | Suggestion.

Appendix C. Work example (Cambridge IELTS 17, Test 4, Band 6, pp. 138, with IELTS examiner’s feedback).

Test 4

WRITING TASK 2

You should spend about 40 minutes on this task.

Write about the following topic:

Nowadays, a growing number of people with health problems are trying alternative medicines and treatments instead of visiting their usual doctor.

Do you think this is a positive or a negative development?

Give reasons for your answer and include any relevant examples from your own knowledge or experience.

Write at least 250 words.

Table 5 Appendix C. Coding Results on B6C17T4

Dimension	Sub-dimension	Score	Justification
Coverage	Coverage	1	The candidate addresses “alternative medicines and treatments” but fails to engage with the required comparative condition “instead of visiting their usual doctor,” resulting in partial fulfillment of the task requirements.
Opinion	Consistency	1 (→2)	A position that “disadvantages outweigh advantages” is stated and maintained. However, the omission of the “doctor” component limits the extent to which this position fully aligns with the task prompt. If evaluated independently of Coverage, internal consistency would merit a 2.
	Clarity	1 (→2)	The opinion is lexically explicit (e.g., “I believe the disadvantages outweigh the advantages”). However, incomplete engagement with the full task scope reduces its functional clarity at the task level, although at the propositional level, the stance would be sufficiently clear to warrant a 2.
Claim	Relevance	1	Main claim (“disadvantages outweigh advantages”) is relevant and clearly articulated. However, the secondary claim concerning the implications of not seeing a doctor is weakly framed and insufficiently supported. Even when Coerage-related penalties are set aside, overall claim quality remains at 1.
	Clear	1	The secondary claim lacks sufficient conceptual clarity, undermining coherence across the argumentative sequence.
Grounds	Connection	1	Arguments (e.g., allergic reactions, environmental benefits, lower cost) are generally related to the stated claims, but explicit logical connections are underdeveloped, particularly in relation to the implications of “not seeing a doctor”.
	Depth	1	Grounds are underdeveloped: examples rely largely on assertions, with minimal elaboration or evidential support. This observation aligns with the examiner’s comment regarding limited depth of development.
Penalty	Word	2	Within word limit.
	Format	2	Formal structure appropriate.
	Repetition	2	No significant redundancy.

Abbreviations

AWE

Automated Writing Evaluation. Technology that uses algorithms to score or provide feedback on written essays, often used for placement testing or practice

CC	Cohesion and Coherence. A scoring criterion in writing assessments (like IELTS) that assesses the flow, structure, and logical connection of ideas within the text
ChatGPT, GPT-4	Generative Pre-trained Transformer
GAI	Generative Artificial Intelligence
IELTS	International English Language Testing System
I-CVI or S-CVI	Item-Level Content Validity Index or Scale-Level Content Validity Index. A metric used instrument development to quantify the degree of agreement among experts regarding the relevance of an individual item, or to quantify the overall content validity of the entire scale or instrument, often calculated as the average of all I-CVI scores
MAE	Mean Absolute Error. A common statistical metric used to measure the average magnitude of errors in a set of predictions, specifically in AWE and machine learning accuracy
PEG	Proprietary Essay Grader. An early and well-known example of an AWE tool developed by Ellis B. Page, used to automatically score essays
TR	Task Response. A scoring criterion in writing assessments (like IELTS) that evaluates how well the writer addresses the prompt and develops relevant ideas
ZPD	Zone of the Proximal Development

Supplementary Information

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Supplementary Material 1.

Authors' contributions

Weiqiang Li: Conceptualization, Data Curation, Formal Analysis, Methodology, Software, Validation, Visualization, Original Draft. Nur Ainil Sulaiman: Supervision, Methodology, Review & Editing. Mohd Effendi Ewan Mohd Matore: Supervision, Methodology, Review & Editing. Xin Ke: Data Curation, Software, Validation.

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Data availability

No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate

(Not applicable)

Consent for publication

(All authors have read and approved the final manuscript for publication)

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The authors declare no competing interests.

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